

SC3102: Signals, systems and transforms

Tutorial 3: Systems

1. Consider the following system outputs. Determine whether the system has following properties: (i) Memory, (ii) Causal, (iii) Time-invariance, and (iv) Linearity

(a) $y(t) = |x(t)|$

(b) $y(t) = x(t - 1)$

2. The goal of sampling is to convert a continuous-time signal $x(t)$ to a discrete-time signal $x[n]$. Ideal sampling can be described mathematically as a product of $x(t)$ with an impulse train:

$$y(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

where T is the sampling interval. Is sampling a linear operation? Is it time-invariant? Is it causal?

3. Step response of a LTI system is the output of the system when the input is a unit step function $u(t)$. Let $g(t) = \Pi\left(\frac{t-1}{2}\right)$ be the step response of some LTI system. Find the output of this system when the input $x(t) = \Pi(t)$. Is this a causal system?

4. Determine the output of each of the following LTI systems with given impulse response $h(t)$ and input $x(t)$.

(a) $h(t) = \delta(t - 1)$, $x(t) = \sin\left(\frac{\pi t}{2}\right)$

(b) $h(t) = u(t)$, $x(t) = (t + 1)u(t + 1)$

(c) $h(t) = u(t) - u(t - 2)$, $x(t) = r(t)u(3 - t)$

5. Are the following systems bounded input-bounded output (BIBO) stable?

(a) $y[n] = \sum_{k=-\infty}^n x[k]$

(b) $y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k]$

6. Let $h(t) = e^{-t}u(t)$ and $H(\omega) = \frac{1}{1+j\omega}$ be the impulse response and the corresponding frequency response of an LTI system. Compute the system output for $x(t) = \cos(t) + \sin(3t)$.

Answers:

1. (a) Memoryless, causal, time-invariant, non-linear
(b) With memory, causal, time-invariant, linear
2. The system is linear, but not time-invariant. It is a causal system.
3. Output of the system = $\Pi\left(\frac{t-\frac{1}{2}}{2}\right) - \Pi\left(\frac{t-\frac{3}{2}}{2}\right)$. It is a causal system.
4. (a) $y(t) = -\cos\left(\frac{\pi t}{2}\right)$
(b) $y(t) = \left(\frac{1}{2}t^2 + t + \frac{1}{2}\right)u(t+1)$
(c) Use graphical approach to find the solution.

$$y(t) = \begin{cases} 0, & t < 0 \\ \frac{t^2}{2}, & 0 \leq t < 2 \\ 2t - 2, & 2 \leq t < 3 \\ \frac{(5-t)(t+1)}{2}, & 3 \leq t < 5 \\ 0 & t \geq 5 \end{cases}$$

5. (a) The system is unstable.
(b) The system is stable.
6. $y(t) = \frac{1}{2(1+j)}e^{jt} + \frac{1}{2(1-j)}e^{-jt} + \frac{1}{2j(1+3j)}e^{j3t} - \frac{1}{2j(1-3j)}e^{-j3t}$

Exercise:

1. Consider the system $\mathcal{H}[x(t)] = a$ where a is a constant. For which values of a is the system linear? For which values of a is the system time-invariant?
2. Which of the systems listed below are linear? Which are time-invariant? Which are causal? If possible, find the impulse response of the system.
 - (a) $\frac{d^2}{dt^2}y(t) + 3\frac{d}{dt}y(t) - ty(t) = 3x(t+2)$
 - (b) $y(t) - x(t) - x(t-4) = 3$
 - (c) $y(t) = x(t) + 3x(t-2) + \int_{t-10}^{t-3} x(\tau)d\tau$
 - (d) $y(t) = \sin(2t)x(t)$
3. A system has the impulse response $h(t) = u(t)$ where $u(t)$ is a unit step function. Using convolution integral, find the system response when the input signal $x(t)$ is
 - (a) $u(t-2)$
 - (b) $e^{-2t}u(t)$
 - (c) $tu(t)$
 - (d) $t^2u(t)$

- (e) Consider $h(t) = \sum_{-\infty}^{\infty} \delta(t - 2n)$, find the system response for an arbitrary input signal $x(t)$
- (f) A system has an impulse response $h(t) = 4e^{-5t}(u(t - 3) - u(t - 6))$, where $u(t)$ is a unit step function. Using graphical approach, obtain the system response corresponding to the input signal $x(t) = \Pi\left(\frac{t-3}{2}\right)$, where $\Pi(\)$ is a rectangular function. Show the steps clearly.

4. An LTI system has an impulse response given by

$$h[n] = \left(\frac{4}{7}\right) 2^{n-1} u[-n] + \frac{11}{7} \left(\frac{1}{4}\right)^{n-1} u[n-1]$$

where $u[n]$ is the unit step function.

- (a) Is the system causal? Justify.
- (b) Is the system bounded input-bounded output stable? Justify.

$$\text{Use } \sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}, \text{ if } a \neq 1.$$

Answers (Exercise):

1. The system is linear only when $\alpha = 0$. The system is time-invariant for any a .
 2.
 - (a) Linear, time-invariant and not causal
 - (b) Non-linear, time-invariant and causal;
Impulse response: $h(t) = \delta(t) + \delta(t - 4) + 3$
 - (c) Linear, time-invariant and causal;
Impulse response: $h(t) = \delta(t) + 3\delta(t - 2) + u(t - 3) - u(t - 10)$
 - (d) Linear, not time-invariant and causal.
 3.
 - (a) $y(t) = (t - 2)u(t - 2)$
 - (b) $y(t) = \frac{1}{2}(1 - e^{-2t})u(t)$
 - (c) $y(t) = \frac{1}{2}t^2 u(t)$
 - (d) $y(t) = \frac{1}{2}t^3 u(t)$
 - (e) $y(t) = \sum_{-\infty}^{\infty} x(t - 2n)$
 - (f)
$$y(t) = \begin{cases} 0, & t < 3 \\ -\frac{4}{5}(e^{-5t} - e^{-15}), & 3 \leq t < 5 \\ -\frac{4}{5}(e^{-5t} - e^{-5(t-2)}), & 5 \leq t < 6 \\ -\frac{4}{5}(e^{-30} - e^{-5(t-2)}), & 6 \leq t < 8 \\ 0, & t \geq 8 \end{cases}$$
 4.
 - (a) The system is non-causal.
 - (b) The system is bounded input-bounded output stable
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